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The latest *in vitro* models for novel uveal melanoma drug discovery: how effective are they and what needs to be done?

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Abstract

Introduction: Uveal melanoma (UM) is the most common primary intraocular malignancy in adults, but primary tumor treatment carries a high risk of permanent vision loss and does not adequately prevent metastatic progression. *In vitro* UM models are needed to accurately represent human disease to support translation of laboratory research to the clinic.

Areas covered: This review covers current and emerging *in vitro* UM models. A PubMed search used keywords ‘uveal melanoma’ and ‘cell line,’ ‘spheroid,’ ‘organoid,’ ‘culture,’ or ‘*in vitro*’ to identify cell lines, three-dimensional (3D) cultures (spheroids and organoids), and co-culture systems. Model successes and shortcomings are described, considering features that make models more or less representative of *in vivo* human UM. Insights are provided for consideration when selecting UM models for novel drug discovery.

Expert opinion: While traditional cell lines have provided an important foundation for UM research, emerging spheroid and patient-derived organoid models may more accurately represent *in vivo* tumor behavior and the tumor microenvironment. Pairing these 3D models with co-culture techniques could dramatically improve the representativeness of UM models. Researchers should consider testing promising therapeutics on a panel of models representing different UM subtypes, with particular attention to high-risk UM, such as those with BAP1 loss.

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Keywords

Cell lines; co-culture systems; *in vitro* models; isogenic cell lines; patient-derived organoids; precision medicine; spheroids; uveal melanoma

1. Introduction

Uveal melanoma (UM) is the most common primary intraocular malignancy occurring in the adult patient population with incidence ranging from <1 to >9 per million persons per year, increasing in frequency for populations with fair skin and light eye color [1–3]. Despite this, treatment options remain limited, and UM continues to have a devastating impact on both patient quality of life and overall survival [4–14]. Standard of care treatment for most primary UM is local radiation, but in an analysis of >1,000 patients, the 4-year Kaplan-Meier estimated risk of visual acuity 20/200 or worse was 61%, leaving more than half of treated patients with severely reduced vision in the treated eye [11]. Following this sacrifice of eyesight, overall survival remains disappointingly poor despite high rates of local tumor control; up to half of all patients diagnosed with primary UM will ultimately develop metastatic disease [13]. Overall treatment response rates for metastatic UM are <20% [4–7], even with modern immunotherapy, and median overall survival (OS) is <2 years [4,7]. Most patients diagnosed with metastatic UM will die of the disease. Thus, current treatments for UM do not adequately preserve vision-related quality of life or improve OS [4–9].

To effectively screen novel therapeutics that may improve visual acuity preservation and OS for patients affected by UM, *in vitro* laboratory model systems are needed that accurately represent *in vivo* therapeutic response. Uveal melanocytes, found in the iris, ciliary body, and choroid, are the presumed cell of origin for UM [15–17], and, importantly, UM is a molecularly distinct cancer from its more common cutaneous counterpart [3,18]. In contrast to cutaneous melanoma, UM has a low mutational burden, limited UV mutational signature (for the more frequent posteriorly located tumors), and lacks pathognomonic cutaneous melanoma mutations in *NRAS* or *BRAF* [3,18–22]. UM also has a unique pattern of metastasis, with the liver as the overwhelmingly most frequent metastatic site [23,24]. These features limit the applicability of targeted anticancer therapeutics for cutaneous melanoma to UM and may in part explain the poor response of UM to traditional immunotherapy [3,18,25]. Furthermore, these distinctions highlight the need to study UM as an independent disease. Most UM (>90%) carry a mutually exclusive mutation in *GNAQ* or *GNA11*, with a small subset of UM being driven by mutations in *CYSLTR2* or *PLCB4* [3]. However, most choroidal nevi also carry a *GNAQ* or *GNA11* mutation [26]. Additional mutations, such as those in *EIF1AX*, *SF3B1*, or *BAP1*, are believed to be necessary for progression to UM and carry implications for metastatic risk [3]. In particular, loss of BAP1 (BRCA1 associated protein 1) protein expression is strongly associated with UM progression to metastasis and death [19,27–30]. Unfortunately, the staggeringly high attrition rate of UM clinical trials [31–38] suggests that historically utilized UM laboratory models may not be adequately representative of human disease [23,39], potentially providing falsely optimistic tumor response results [23,40–43].

To add complexity to the need for models, as with other cancers, UM is not a homogenous disease. There are distinct UM subtypes with unique genomic and transcriptomic profiles. Efforts through The Cancer Genome Atlas (TCGA) identified prognostically significant chromosomal alterations, with higher metastatic risk associated with monosomy 3 and 8q gains [19]. Analysis of gene expression profiles has stratified UM into lower risk Class 1 and higher risk Class 2 tumors, with preferentially expressed antigen in melanoma (PRAME) positive tumors have further increased metastatic risk [44,45]. While the prognostic implications of these differences have been well-defined, the critical determinants of therapeutic response are less well-understood. To represent UM subtypes and their potentially unique therapeutic profiles in the laboratory, a variety of models are required to support personalized *in vitro* drug response studies.

In this review, we discuss the evolution of *in vitro* UM models to support novel UM drug discovery. We explore commercially available cell lines, including their strengths and shortcomings and discuss novel models such as spheroids and patient-derived organoids (PDOs) that may better represent the spectrum of human disease and personalized UM therapeutic sensitivities.

A literature search was undertaken in November 2025 using PubMed for English language articles describing UM models published between 1966 through the time of search. Keywords in the literature search included ‘uveal melanoma’ paired with ‘cell line,’ ‘spheroid,’ ‘organoid,’ ‘culture,’ or ‘in vitro.’ Included model types for this review were cell lines, three-dimensional (3D) cultures (spheroids and patient-derived organoids), and co-culture systems. A flowchart depicting the cultivation process for different types of model systems is shown in Figure 1.

2. Traditional uveal melanoma cell lines

The development of UM cell lines has offered an important opportunity to study UM behavior in the laboratory setting. However, some issues have arisen throughout cell line development which have required refinement to ensure validity of laboratory research. In 1989, two cell lines were described, OCM-1 and OCM-2 [46], which were reportedly established from biopsied choroidal melanoma. However, OCM-1, along with other reported UM cell lines OCM-3, OCM-8, MUM2C, MKT-BR, and SP-6.5, were found to have a mutation in *BRAF* [47–51]. Since the time of those reports, studies have shown that, unlike cutaneous melanoma, UM lacks mutations in *BRAF*, suggesting that those cell lines may actually be cutaneous in origin rather than uveal [3,18–22]. This could have occurred due to laboratory contamination events or due to samples taken from a patient with both uveal and cutaneous melanoma, where the tissue origin of a sample was in fact cutaneous but clinically interpreted as uveal. Alternatively, if the patients from whom these samples were derived were identified as having ‘ocular melanoma’ without specificity to the ocular site of origin, it is possible that such tumors were conjunctival melanoma, which has a molecular profile more similar to that of cutaneous melanoma [52]. Unfortunately, studies done on such cell lines may not have accurately represented UM behavior or therapeutic sensitivity in the laboratory.

The origin of some UM cell lines has also been drawn into question due to a lack of melanocyte markers. Less consistent with their reported UM origin, Mel285 and Mel290 lack the expression of either HMB45 or Melan-A [53]. The primary tumors from which these cell lines were derived were highly positive for Melan-A/MART-1 and HMB-45, which suggests that perhaps a different cell population, such as tumor-associated fibroblasts, could have taken over these cultures [49]. These concerns are further purported by a lack of mutations in either of the two main drivers of UM, *GNA11* and *GNAQ*, in these samples [49]. Multiomics analyses suggested that these two cell lines are distinct from other UM lines [54]. Therefore, these cell lines may not be recommended for use as representative UM models.

As with cell culture of any cell type, laboratory contamination events or sample mixups can occur, which could explain why some cell lines were shown to be likely the same line after short tandem repeat (STR) testing. Cell lines that appear to share the same background, or at least be derived from the same patient, include OCM-1 with MUM2C, OCM-3 with OCM-8, and M619 with both C918 and MUM2B. Further, OCM-3 and OCM-8 were highly similar to SK-Mel28, which is a well-known cutaneous cell line, providing additional support to suspicions of their cutaneous origin based on the presence of a *BRAF* mutation [48]. Interestingly, while the cell line 92.1 has been broadly used, two different STR profiles have been published for that cell line, which could indicate genetic drift in cell culture or other issues such as laboratory contamination events. While 92.1 does appear to be a true UM cell line even in the presence of different STR results, genetic drift in cell culture could potentially impact experimental reproducibility [48,49]. As with any cell line development, thorough validation, including STR testing, is critical, with repeat STR testing at times of laboratory transfers and periodically during routine experimental use. Ideally, analysis should compare key features of the primary tumor with those in the cell line, including immunohistochemical markers and pathognomonic mutations, to be sure the cell line retains characteristic features that may drive tumor behavior and therapeutic response.

After considering whether the origin of a cell line model was truly UM and accounting for only those cell lines with unique identities, another important factor to consider is whether or not UM cell lines accurately represent most UM tumors. Indeed, clinical histories of patients from whom cell lines have been derived may not be typical of most patients with UM, suggesting that some cell lines may be derived from rare or unusual UM subtypes [49]. The cell line 92.1 was derived from a patient with a massive UM with extraocular extension, requiring exenteration, and the tumor led to widespread metastases [55,56]. The cell line has a mutation in *EIF1AX*, although this has not been confirmed in the primary tumor, and the cell line has disomy chromosome 3, retained *BAP1* expression, and an absence of *BAP1* mutations [49]. This is somewhat less common, as most UM that metastasize have *BAP1* loss. The cell lines Mel202 and Mel270 were derived from ocular recurrences following initial primary tumor irradiation, so they represent cells that either recurred following irradiation or growth of a population of cells that survived initial radiation exposure [49]. Mel290 was derived from a tumor arising in the setting of oculodermal melanocytosis, which is a known risk factor for developing UM but is rare, impacting on 1–2% of UM cases [57–59]. Mel290 lacks mutations in *GNAQ*, *GNA11*, *SF3B1*, (exon 12, 13, 14, 15), *EIF1AX*, or *BAP1* and does not show loss of heterozygosity at chromosome 3, although

a lack of 3p26-pter and gain of 8q24.1-q24.2 has been reported [48,49]. The primary tumor of origin has chromosome 3 monosomy in 80% of cells [49], so this cell line might not represent the majority of cells within that tumor. Mel285 was derived from a typical ciliochoroidal melanoma, but the cell line lacks expression of melanocyte markers [49]. Further, while the original tumor had a *GNA11* Q209 mutation, that mutation was not identified in the cell line [48]. Therefore, many UM cell lines, while useful, may not be accurately representative of typical human UM. Understanding the clinical history of donor patients is important to validate the translational relevance of a cell line, and even when derived from a more typical tumor, the cell line may not match the genotypic profile of the clinical specimen.

UM cell lines have been generated from both primary and metastatic UM tumors [60]. In some instances, metastatic cell lines have been generated from the same patients as primary UM cell lines, which may offer an important opportunity to compare primary and metastatic UM behavior. For example, OMM2.3, and OMM2.5 were generated from two metastases of the patient who donated primary tumor tissue for Mel270 [48]. However, despite the metastatic behavior in the patient, these cell lines had chromosome 3 disomy and retained BAP1 expression, which may be less representative of the majority of metastases and could represent outgrowths of selective cell populations from a heterogeneous tumor [49]. In fact, the patient's primary tumor had loss of one chromosome 3 and loss of BAP1 expression, so the cell line does not appear to accurately replicate *in vivo* tumor features [49].

Generating cell lines from primary UM has been notoriously challenging. Poor plating efficiency and long doubling times exceeding 400 hours in some reports can prove difficult to overcome [61]. One group generated two cell lines, MP38 and MP65, directly from primary human UM tumors and reported their success rate for cell line generation at only 3% [62]. The same group had additional success generating cell lines from patient-derived xenograft (PDX) models of primary (MP41 and MP46) or metastatic (MM28, MM33, MM66) UM [62,63]. Another group reported successful UM cell line establishment in less than 5% of attempts [64]. Transient primary cell line cultures offer some appeal in potential maintenance of tumor heterogeneity and molecular markers [65,66], but the short-term nature of such cultures may preclude experimental reproducibility. The challenges generating lasting primary cell lines may in part explain the paucity of UM cell lines with BAP1 loss [48,62,67,68]. Despite these difficulties, given that loss of BAP1 expression such a strong predictor of metastatic risk [19,27], studying *in vitro* models that represent this feature is paramount to making progress toward improved understanding of UM pathobiology and development of effective treatment strategies. A summary of known clinical origin and molecular features of unique UM cell lines is provided in Table 1.

3. Uveal melanoma spheroids

Even after improved validation of UM cell lines, such models retain inherent shortcomings that may limit translation of experimental results. Therefore, researchers have sought opportunities to develop more representative model systems that may more accurately replicate *in vivo* pathobiology of UM. While two-dimensional growth does not accurately model tumor biochemical and mechanical signals, tissue architecture, or cell to cell and

cell to extracellular matrix communications, three-dimensional (3D) models may offer an opportunity to better study these features and may also more accurately represent UM *in vivo* drug response [23,69,70].

Spheroids can be grown from both commercial UM cell lines and primary UM samples, with primary UM spheroids retaining genetic features of the primary tumor [71]. When grown as spheroids, an outer layer of viable, proliferating cells may be observed, with a less active, sometimes hollow or apoptotic central core [72]. Both anchorage-dependent and anchorage-free-based methods have been used to grow 3D UM cultures [72–75]. Anchorage-dependent methods involve seeding or embedding cells in a platform of proteins simulating extracellular matrix [76], while anchorage-free methods, also known as the multicellular tumor spheroid (MCTS) model, involve single cell suspensions of cancer cells in serum-supplemented media, preventing cell adherence and simulating the heterogeneity of abnormal, vascularized tumors [77,78]. Using anchorage-free methods, commercial UM cell lines formed multicellular tumor structures with the outer layer of cells having higher vitality and proliferation, simulating an abnormal, vascularized solid tumor [72]. Under anchorage-dependent conditions, commercial UM cells had variable morphology depending on the cell line, with some forming flattened structures and other forming tight clusters [72]. Some cells were unable to form cell-to-cell interactions in 3D culture [72]. Cells with an epithelioid morphology appeared to have more invasive potential than those with spindle-like morphology, which is consistent with clinical knowledge that epithelioid tumors may be more aggressive [72]. Interestingly, regardless of the method used, UM grown in 3D cultures released more vascular endothelial growth factor in response to hypoxia than UM grown in two dimensions [72]. When grown in 3D, UM cells may require time to reach a steady state of compaction prior to drug testing, and a combination of different assays may be necessary to evaluate drug response, permitting more representative evaluation of different zones within the 3D structures [71,72].

Other interesting features have been reported when trying to grow UM cell lines in 3D cultures. When plated on Matrigel, aggressive cell lines formed loops and networks [79]. Culturing OCM1 spheroids on non-adherent plates demonstrated migration and phenotypic transformation characteristic of cancer stem cells [80], and other studies have also supported the presence of a cancer stem cell-like subpopulation in UM cell lines that may more accurately simulate *in vivo* UM proliferative and self-renewal capabilities, implicating them as a potentially more representative tool for assessing UM therapeutic sensitivity [81]. Indeed, 3D UM models display different drug response than their 2D counterparts.⁷¹ C198 cells grown as spheroids were found to be more treatment resistant to luteolin than the two-dimensional counterparts, suggesting that 3D models may be a valuable tool in UM drug screening [82]. Other 3D spheroids of UM cell lines have been grown to test novel electrochemotherapy [83,84]. Decitabine has been tested in 3D UM spheroids as a potential method to improve response to MEK inhibitors [85]. Another study found potential therapeutic efficacy of nebulivolol [86]. Migration may also be more accurately studied using 3D cultures, including assessment of angiogenesis [87].

When comparing gene expression profiles of MCTS models to primary UM tumor biopsies and adherent UM cell *in vitro* models generated from the same patients, principal

component analysis (PCA) revealed that the models clustered based on their culture conditions [75]. The MCTS models demonstrated resistance to anoikis (apoptosis induced by inappropriate or lost cell adhesion), metabolic shift toward a lipogenic profile, and upregulation of *synovial sarcoma X breakpoint (SSX)* genes, which have been associated with anoikis resistance and epithelial-mesenchymal transition [75]. While these changes could suggest potential mechanisms driving UM dissemination, additional research is required to understand if these changes accurately represent what occurs with *in vivo* UM metastatic progression.

4. Uveal melanoma patient-derived organoids

Patient-derived organoids (PDOs) have been a useful tool for drug screening in a wide variety of other cancers. In some cancers with more effective treatments, PDO drug response has demonstrated good correlation to clinical outcomes in corresponding PDO donor patients [88]. Compared to two-dimensional cultures, PDOs may better replicate the 3D tumor microenvironment and 3D tissue functions of *in vivo* tumors [49,62,89,90]. PDOs may also better maintain intratumoral heterogeneity representative of the *in vivo* tumor as opposed to cell lines that have outgrowth of one particular cell type [49,62,89,90]. These models overcome ethical concerns associated with *in vivo* models and may be a more rapid, renewable resource for high throughput drug screening [49,62,89–91]. Importantly, in the Food and Drug Administration (FDA) Modernization Act 2.0 of 2022, the US FDA endorsed organoid models as a new alternative method for drug development that could circumvent the need for animal testing [92].

A number of different culture techniques have been described to utilize PDOs in other cancers, including cultures in extracellular matrices, air-liquid interfaces, microfluidics, and organ-on-chip [90]. However, PDOs have been a more recent addition to the repertoire of UM models. In UM, a biobank of UM PDOs has been developed via culturing freshly dissociated tumor samples on Cultrex (R&D Systems 3533–005–02) (Figures 2 and 3) [91]. At least 40 unique PDOs have been established, and the PDOs accurately replicate key features of the corresponding primary tumors, including UM driver and prognostic mutations and gene expression profiles [91]. The PDOs have more representative drug resistance to agents that have previously failed in UM clinical trials, such as MEK inhibitors, and the PDOs show differential drug response, suggesting their utility to support precision medicine studies for personalized drug selection [91]. Importantly, the PDO biobank contains models representing a wide variety of UM phenotypes, which provides an important foundation for future personalized medicine studies [91]. Indeed, PDOs have been generated to test novel combination therapies. In one study, PDOs demonstrated translational promise for combination treatment with SRC inhibitors and autophagy-inducing drugs for cancers with BAP1 loss, including UM [93]. This early work suggests PDOs are promising models for UM, but future work is needed to develop best practices for PDO generation and propagation unique to this cancer type.

5. Isogenic cell lines

Paramount to understanding mechanisms of UM pathobiology is the ability to study isogenic cell lines. However, the presumed origin cell for UM, the uveal melanocyte, is not commercially available. Protocols have been developed for the generation of choroidal melanocyte cell lines from donor eyes, and the generation of such cell lines has allowed important comparisons of the transcriptome and proteome of UM compared to its progenitor cell [17,94]. Beyond this, to study the impact of a critical UM driver mutation in isolation, *GNAQ*^{Q209L} mutant choroidal melanocyte cell lines have been developed [94]. Normal uveal melanocytes can also be grown as spheroids when seeded at high density on a matrix-like substance such as Matrigel™ (Corning) or Cultrex (R&D Systems 3533–005–02) [17,94]. Future studies could build upon this work to better understand the mechanistic contributions of known UM mutations using isogenic cell line models, including additional models with engineered mutations in *GNA11*^{Q209L}, *EIF1AX*, *SF3B1*, and *BAP1*. However, variability in cell viability and lack of immortal behavior, resulting in transient cell culture life, continue to pose challenges to using normal human uveal melanocytes as a research platform [17,94]. Immortalization of uveal melanocyte cell lines could be helpful in the future, although validation would be required to ensure lack of meaningful differences in cell behavior following immortalization.

6. Co-culture systems

Co-culture model systems may be useful for studying the interaction between cancer cells and the surrounding tumor microenvironment. While such studies have been limited in UM, early work may provide insight into potential future applications of co-culture models. Using Transwell inserts with 1 µm pore size, 92.1 cells have been co-cultured for 6 days to condition human retinal pericytes via mutually produced diffusible factors, revealing UM-induced morphological changes in the pericytes and increased UM invasion, suggesting that the pericytes attained a cancer-activated fibroblast phenotype [95]. Using a similar 1 µm pore size Transwell insert method to permit exposure to diffusible factors without allowing cell migration, Mel270 and OMM2.3 have been co-cultured with LX-2 activated hepatic stellate cells. Paracrine signaling of hepatic stellate cells resulted in upregulation of inflammation-associated transcripts, with greater response of metastatic UM compared to primary UM cells [96]. These early studies suggest that co-culture methods can be used to better understand how UM and the cells in the surrounding tumor microenvironment influence one another's behavior both within the intraocular primary tumor environment and in the metastatic setting, particularly the liver as the most frequent metastatic site. Such methods could be used to better understand factors driving metastasis and organotropism. Future co-culture studies using more representative 3D UM model systems may be warranted to better understand the relationship of UM with its microenvironment in both the primary and metastatic setting.

More tissue-like model systems may be of interest in the future as well [97]. A tissue-engineered choroid model was developed that includes choroidal melanocytes, fibroblasts, endothelial cells, retinal pigment epithelial cells, and extracellular matrix [98]. The retinal pigment epithelium/choroid complex was dissected off the sclera of human donor eyes,

and retinal pigment epithelium cells were dislodged for separate culture. Choroidal vascular endothelial cells were extracted from a dissociated cell suspension for culture. Washing fractions collected during endothelial cell isolation were split in half to separately culture choroidal stromal fibroblasts and choroidal melanocytes. Choroidal stromal fibroblasts were seeded using a paper anchoring ring, forming matrix sheets after 6 weeks of culture. Two sheets were superimposed for increased thickness. The tissue engineered choroidal stromas could then be seeded with choroidal melanocytes, endothelial cells, and retinal pigment epithelial cells. This model may also be seeded with UM spheroids to permit *in vitro* study of the tumor microenvironment in primary UM [99]. Future improvements in methods to replicate the liver extracellular matrix would prove useful for studying UM in early and late stages of metastatic progression [97].

7. Conclusion

There are a wide range of *in vitro* UM models with varying ability to accurately represent *in vivo* human disease. Two-dimensional UM cell lines have been widely used but in many cases have lacked key features of UM, such as UM driver mutations, melanocyte markers, or chromosomal alterations found in the primary tumor. Successfully generated cell lines were often derived from clinically atypical tumors that may not represent most UM, and high-risk UM with BAP1 loss has proven challenging to culture. Newer 3D culture systems, including spheroids and patient-derived organoids, may better represent *in vivo* UM pathobiology, including maintenance of tumor heterogeneity and tumor microenvironment. Isogenic cell lines and co-culture systems may prove advantageous for studying targeted molecular or pathophysiologic questions. When considering UM model use for drug discovery studies, it is important to be aware of the strengths and limitations of the chosen model system to understand the likelihood for translational success.

8. Expert opinion

Ensuring the representative nature of *in vitro* UM models remains a challenge, but doing so is essential to producing results that will successfully translate to the clinical setting. While cell line models are easy to work with and may permit the most rapid drug screening, numerous failed clinical trials based on commercial UM cell line studies suggest that these models may have more optimistic drug sensitivities than *in vivo* human tumors. Commercially available cell lines have been fraught with issues that may result in lack of representative drug response, including genetic drift in two-dimensional cell culture, poor retention of chromosomal alterations and genetic mutations compared to the tumor of origin, lack of melanocyte markers, and derivation from clinically atypical tumors that may not represent most patients with UM [49]. Furthermore, a large body of work has been done on UM models with retained BAP1 expression, but current literature supports BAP1 loss as one of the strongest phenotypic predictors of UM-related metastasis and death [19,27–30]. To support drug discovery for agents that will successfully prevent metastatic progression in a majority of patients with UM, researchers must study UM models with BAP1 loss. Ideally, some models would be derived from UM tumors with native BAP1 loss rather than artificial BAP1 knockdown because BAP1 retained cell lines with inducible BAP1 knockdown [40,42] may lack important additional drivers of UM behavior. This

possibility appears to be suggested by studies of genetically engineered mouse models in which BAP1 loss had no impact on UM tumor progression [42,100]. Therefore, studying UM with molecular features representative of high metastatic risk may improve the chances of successful clinical translation.

Creating isogenic cell lines to study specific molecular alterations appears feasible and may provide a model system best suited to probe the functional and mechanistic consequences of targeted mutations. However, these models may lack the concomitant alternations seen in UM tumors, making them less ideal for drug discovery.

Overcoming some shortcomings of traditional UM cells lines, the emergence of 3D UM models, such as spheroids and patient-derived organoids, offers promise for improved retention of primary tumor features, including better representation of the tumor microenvironment, intratumoral heterogeneity, and cellular interactions that may only occur in 3D tissue. Such models may have more realistic drug response compared to two-dimensional cell lines, with enhanced treatment resistance to agents that have previously failed in clinical trial [91]. Still, these models are only as good as the tissue or cells from which they were derived. Spheroids derived from commercial cell lines will carry inherent limitations if the chosen cell line has atypical features. Patient-derived organoids may overcome this issue, but just as with two-dimensional models, care must be taken to adequately validate retention of primary tumor features both during initial model establishment and over time with increasing duration in culture. Future work is needed to determine how well these 3D models retain key features over long duration culture and transfer between different laboratories.

Co-culture systems offer a promising pathway toward better understanding tumor function within the tumor microenvironment. Further work is needed to develop platforms that reliably replicate areas of interest such as the liver extracellular matrix. However, combining a co-culture strategy with a 3D UM model could provide a more effective way to represent UM behavior in the lab.

With so many different groups developing different methodology and different cell lines and 3D cultures, there is a need for better collaboration and coordination to establish consistent methodology and UM model characterization criteria that will support more effective translation of UM laboratory research to the clinic. Although it is tempting to look for the single most ideal model, the search should perhaps be more focused on an ideal model system. Within the ideal model system, researchers should strongly consider testing any new promising drug in multiple UM models with distinct genotypes to provide data on which specific patients are most likely to benefit from treatment. It is important to remember that UM is not a homogenous disease. Although it is a rare cancer, with the advent of patient-derived organoid biobanks, the resources are now available to bring UM research into the future with personalized medicine.

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Article highlights

- Uveal melanoma *in vitro* models may include cell lines, spheroids, patient-derived organoids, and co-culture systems.
- Careful model validation is necessary, as some traditional cell lines have been found to lack characteristic uveal melanoma mutations or melanocyte markers.
- Consideration should be given to the origin of model systems, as some may have been derived from less representative, clinically atypical tumors.
- Incorporation of models with high-risk features, such as BAP1 loss, is important when considering *in vitro* validation of novel therapeutics.
- Three-dimensional models, such as spheroids and patient-derived organoids may be more representative of *in vivo* tumor heterogeneity and interactions between cells and their microenvironment.
- Co-culture systems may offer a future opportunity to further improve *in vitro* representation of *in vivo* tumor pathobiology.

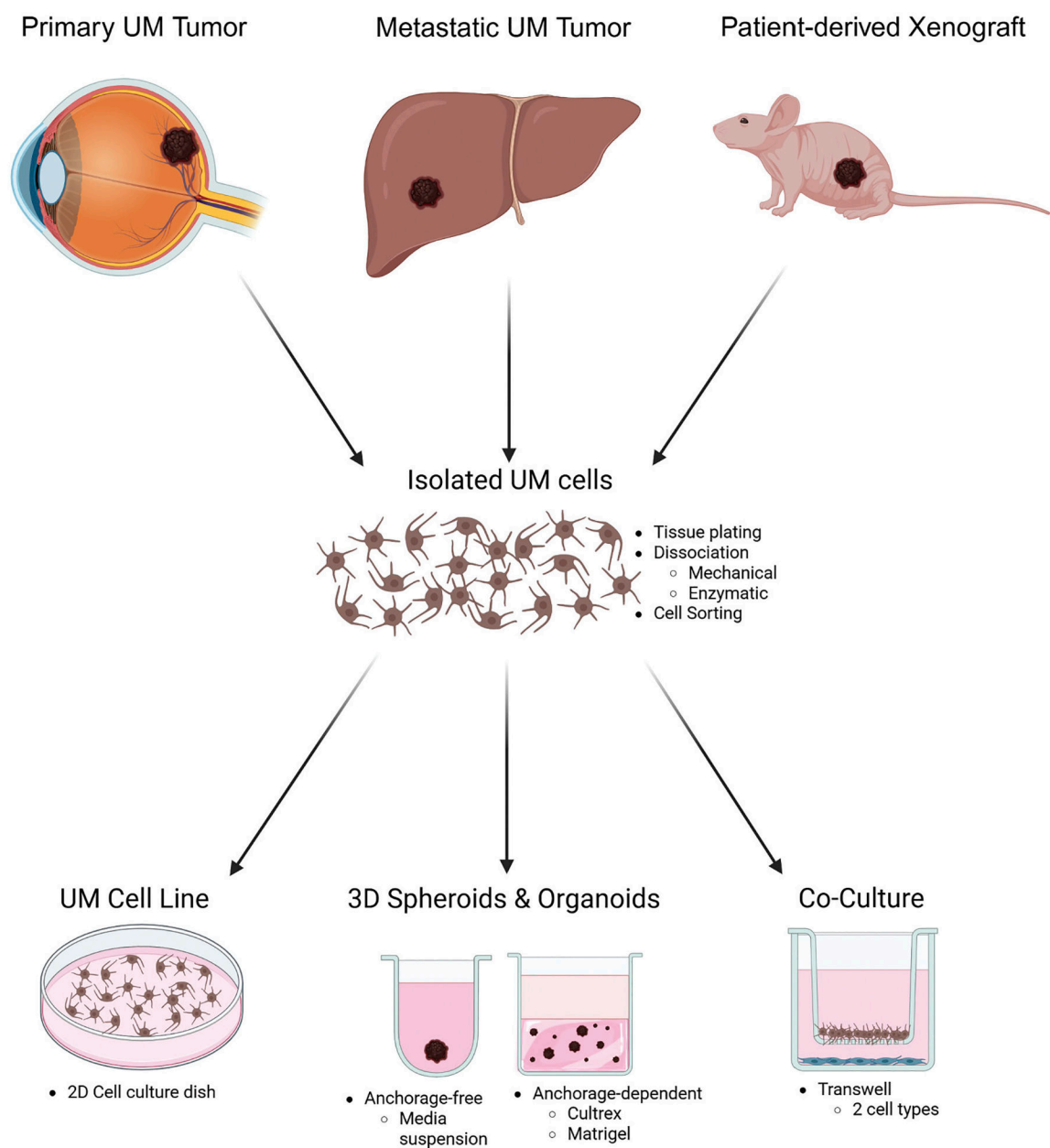


Figure 1.

Cultivation process for distinct uveal melanoma model systems. A schematic depicts the general process for developing uveal melanoma cell lines, three-dimensional cultures such as spheroids or organoids, and co-culture systems.

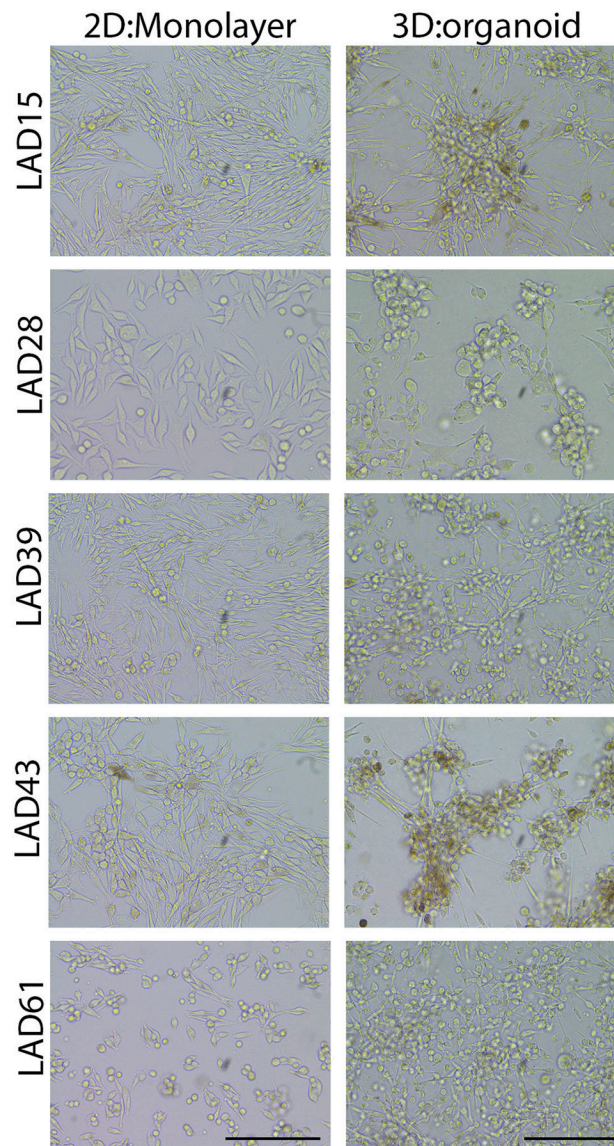


Figure 2.

Patient-derived organoids (PDOs) of uveal melanoma exhibit different organization and morphological characteristics when cultured in three-dimensional (3D) medium compared to two-dimensional (2D) conditions. Distinct PDOs derived from different primary tumors with varying BAP1 expression and genetic mutations were cultured on cell culture-treated plastic dishes or embedded in 3D Cultrex media. Loss of BAP1 expression in primary uveal melanoma is associated with higher metastatic risk, and cultivation of models with different genetic driver and prognostic mutations may allow for studying more personalized mechanistic and therapeutic implications of uveal melanoma subtypes. PDOs cultured in 2D environment grow in a random monolayer distribution pattern with minimal cellular interactions with neighboring cells. In contrast, PDOs in 3D Cultrex medium organize into organoid structures connected by a cellular network. The 3D culture environment allows for

a more accurate representation of the *in vivo* microenvironment compared to a 2D *in vitro* environment. Images were taken with a Nikon DS camera with scale bar equal to 200 μ M.

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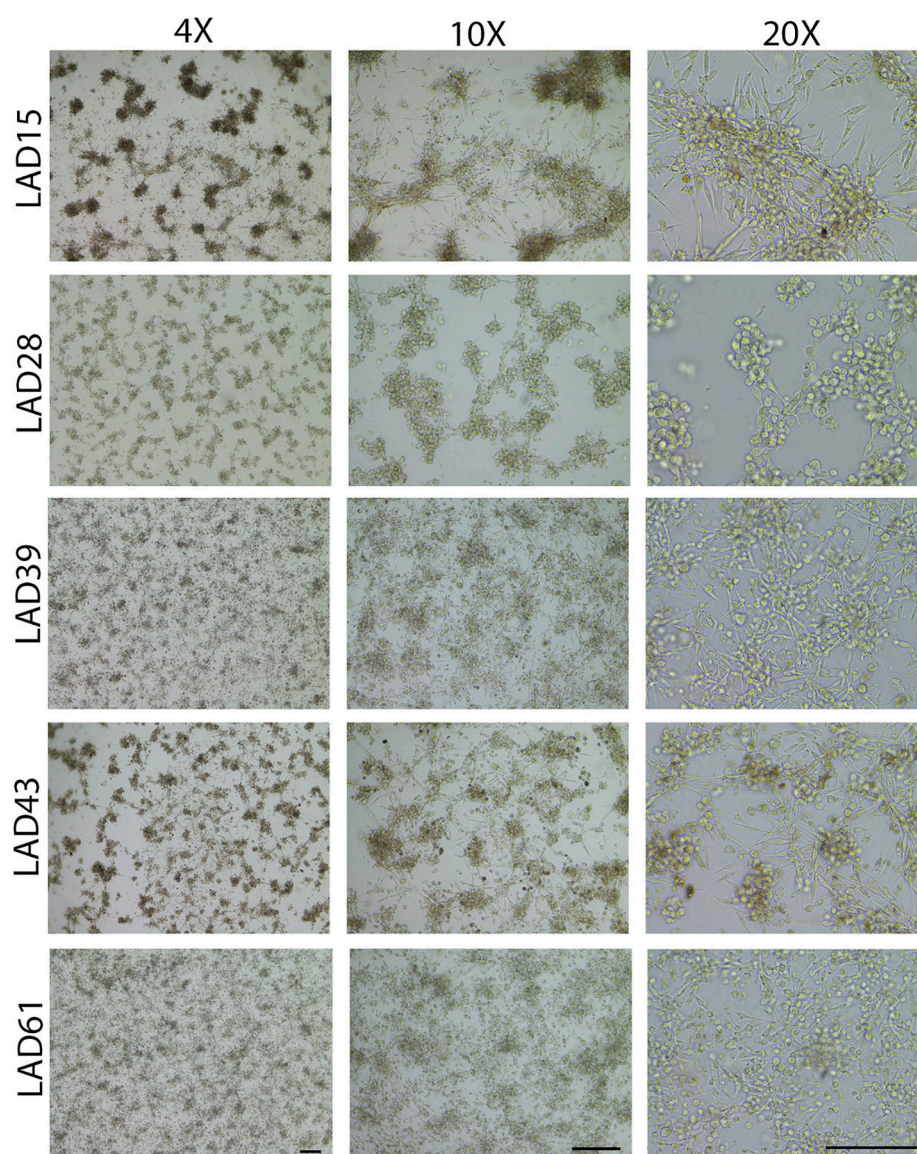


Figure 3. Patient-derived organoids (PDOs) of uveal melanoma cultured in three-dimensional (3D) Cultrex medium form organoid structures. Multiple PDOs with varying genetic mutations, including models derived from tumors having high (e.g. BAP1 loss) or low (e.g. BAP1 retained) metastatic risk, were embedded and cultured in 3D Cultrex medium. Each PDO formed organoid structures with varying organization, size, and density of organoid structures. Images were taken with a Nikon DS camera with scale bar equal to 200 μ M.

Table 1.

Overview of authenticating features of *In Vitro* uveal melanoma models.

Model number	Sex	Derived	Origin	Cell Type	BAP1 Protein expression	Driver and Prognostic Mutations					Additional Considerations
						GNA11	GNAQ	CYSLTR2	EIF1AX	SF3B1	
MM28	M	Xenograft	Liver Met	Mixed	Lost	626A > T	–	–	–	–	1881C > A
MM33	N/A	Xenograft	Skin Met	S	Retained	–	626A > C	–	–	–	–
MP38	M	Patient	Primary	S	Lost	–	626A > T	–	–	–	68-9_72del
MP41	F	Xenograft	Primary	Mixed	Retained	626A > A/T	–	–	–	–	–
MP46	M	Xenograft	Primary	Mixed	Lost	–	626A > T	–	–	–	–
MP65	F	Patient	Primary	S	Lost	626A > T	–	–	–	–	1717del
MM66	N/A	Xenograft	Liver Met	Mixed	Retained	626A > T	–	–	–	–	–
MEL202	F	Patient	Primary	Mixed	Retained	629 G > A	–	–	–	–	Melan-A not determined; recurrence following initial primary tumor irradiation
MEL270	M	Patient	Primary	S	Loss	–	626A > C	–	–	–	Recurrence following initial primary tumor irradiation
MEL285	F	Patient	Primary	S	Retained	–	–	–	–	–	Melan-A not detected; mutation present in tumor but not identified in cell line
MEL290	F	Primary	Primary	E	Retained	–	–	–	–	–	Melan-A not detected
OMM1	M	Patient	Subcutis Met	Mixed	Retained	626A > T	–	–	–	–	–
OMM2.3	–	Patient	Liver Met	–	Retained	–	626A > C	–	–	–	Derived from BAP1 retained metastasis in a patient with a BAP1 loss primary tumor
OMM2.5	M	Patient	Liver Met	Mixed	Retained	–	626A > C	–	–	–	Derived from BAP1 retained metastasis in a patient with a BAP1 loss primary tumor
92.1	F	Patient	Primary	Mixed	Retained	–	626A > T	–	17 G > A	–	Two different STR results; mutations

Model number	Sex	Derived	Origin	Cell Type	BAP1 Protein expression	Driver and Prognostic Mutations					Additional Considerations
						GNA11	GNAQ	CYSLTR2	EIF1AX	SF3B1	
M619 (MUM2B, C918)	F	Patient	Primary	E	Lost	-	626A > T	-	-	-	-
LAD11	M	Patient	Primary	Mixed	Lost	-	626A > C	-	-	-	-
LAD15	F	Patient	Primary	S	Retained	-	-	-	-	-	-
LAD17	F	Patient	Primary	E	Lost	-	-	-	-	-	-
LAD25	M	Patient	Primary	Mixed	Lost	626A > T	-	-	-	-	178C > T
LAD28	M	Patient	Primary	S	Retained	-	626A > C	-	17 G > A	-	-
LAD29	M	Patient	Primary	Mixed	Retained	-	626A > C	-	-	-	-
LAD30	F	Patient	Primary	S	Retained	-	-	-	-	-	-
LAD31	M	Patient	Primary	Mixed	Lost	-	-	-	-	-	Recurrence after prior plaque radiotherapy
LAD32	M	Patient	Primary	Mixed	Lost	626A > T	-	-	-	-	136_141del
LAD33	M	Patient	Primary	Mixed	Lost	626A > T	-	-	-	-	1186_1187del
LAD34	F	Patient	Primary	Mixed	Retained	-	-	-	-	1874 G > A	-
LAD36	M	Patient	Primary	Mixed	Lost	-	625_626 CA > TT	-	-	-	442 G > T
LAD37	F	Patient	Primary	E	Lost	626A > T	-	-	-	-	1182C > G
LAD38	M	Patient	Primary	S	Retained	-	-	-	-	-	-
LAD39	F	Patient	Primary	Mixed	Lost	-	-	386T > A	-	-	386T > A
LAD40	M	Patient	Primary	E	Lost	626A > T	-	-	-	-	757C > T
LAD41	M	Patient	Primary	Mixed	Retained	626A > T	-	-	-	1873C > T	-
LAD43	M	Patient	Primary	S	Retained	-	626A > C	-	26 G > A	-	-
LAD44	M	Patient	Primary	S	Lost	626A > C	626A > C	-	-	-	68-2A > C
LAD46	F	Patient	Primary	S	Retained	626A > T	-	-	-	1874 G > A	-
LAD48	F	Patient	Primary	Mixed	Lost	626A > T	-	-	-	-	189_190insT
LAD49	F	Patient	Primary	Mixed	Lost	626A > C	-	-	-	-	758delA
LAD51	M	Patient	Primary	S	Lost	626A > T	-	-	-	-	721_777del 605 G > A

identified in cell line but not tumor

Model number	Sex	Derived	Origin	Cell Type	BAP1 Protein expression	Driver and Prognostic Mutations					Additional Considerations
						GNA11	GNAQ	CYSLTR2	EIF1AX	SF3B1	
LAD52	F	Patient	Primary	Mixed	Retained	626A > T	–	–	–	1874 G > A	–
LAD53	F	Patient	Primary	S	Retained	626A > T	–	–	–	–	–
LAD55	M	Patient	Primary	S	Lost	626A > T	–	–	–	–	1729 G > C
LAD57	M	Patient	Primary	S	Lost	–	626A > T	–	–	–	2128_2162del
LAD58	F	Patient	Primary	Mixed	Lost	508T > A	–	–	–	–	592_597 GAGGAC > AGGAT
LAD59	M	Patient	Primary	Mixed	Lost	–	–	–	–	–	–
LAD60	M	Patient	Primary	Mixed	Lost	–	–	–	–	–	–
LAD61	F	Patient	Primary	Mixed	Lost	–	626A > C	–	–	–	661_714del

Abbreviations: *M* = male, *F* = female, *N/A* = not available, *Met* = metastasis, *Primary* = primary tumor, *S* = spindle, *E* = epithelioid, *ND* = Not Determined.

A dash is included where detected mutations were absent, not reported, or not yet tested.